CSCC37

2025-01-11

1. Floating Point Arithmetic

1.1. Real Number Representation

Definition 11A. For **base** $b \in \mathbb{Z}^+$, a natural $x \in \mathbb{N}$ is represented as

$$x = (d_n d_{n-1} \cdots d_2 d_1 d_0)_b = d_n \cdot b^n + \cdots + d_0 \cdot b^0$$

where $d_k \in \mathbb{Z}_b$ are **digits**.

Algorithm 11A. Convert $N \in \mathbb{N}$ to binary.

1. Construct *numerators* n_k and *remainders* r_k by
   1. Start with $n_1 = N$.
   2. Let r_k be remainder of $n_k/2$.
   3. Let n_{k+1} be quotient (floor) of $n_k/2$.
2. Then

$$N = (\cdots r_3 r_2 r_1)_2$$

Example 11A.

iteration	n	r
1	43	1
2	21	1
3	10	0
4	5	1
5	2	0
6	1	1

$(43)_{10} \mapsto (101011)_2$

Definition 11B. For base $b \in \mathbb{Z}^+$, a real $x \in \mathbb{R}$ is represeted as

$$\begin{aligned} x &= \pm(x_I.x_F)_b \\ &= \pm(d_n \cdots d_1 d_0 . d_{-1} d_{-2} \cdots) \\ &= \pm\left(\sum_{k=0}^{n} d_k \cdot b^k + \sum_{k=1}^{\infty} d_{-k} \cdot b^{-k}\right) \end{aligned}$$

where x_I is the **integral** part, and x_F is the **fractional** part.

Remark 11A. A finite decimal number has finite binary representation. Converse is false, e.g. $(.1)_{10} = (.0\overline{0011})_2$.

Algorithm 11B. Convert fractional $x \in (0,1)$ to binary.

1. Construct *multipliers* m_k and *integrals* i_k by
   1. Start with $m_1 = x$.
   2. Let i_k be integral part of $2m_k$.
   3. Let m_{k+1} be fractional part of $2m_k$.
2. Then
$$x = (.i_1 i_2 \cdots)_2$$

Example 11B.

iteration	m	i
1	.625	1
2	.25	0
3	.5	1

$(.625)_{10} \mapsto (.101)_2$

1.2. Mantissa and Exponent

Definition 12C. Given base b and $t, s \in \mathbb{Z}^+$, the set of **floating point number** (FPN) in base b with **mantissa** t and exponent s is
$$\begin{aligned}\mathbb{R}_b(t,s) &= \{\pm F \times b^e : F \in [0,1), e \in \mathbb{N}\} \\ &= \left\{\pm(.d_1 d_2 \cdots d_t)_b \times b^{(c_1 c_2 \cdots c_s)_b} : d_k, c_k \in \mathbb{Z}_b\right\}\end{aligned}$$
where $(d_1 d_2 \cdots d_t)_b$ is the **mantissa**, and $(c_1 c_2 \cdots c_s)_b$ is the **exponent**.

Definition 12B. An nonzero FPN $\hat{x} = (d_1, \cdots, d_t, c_1, \cdots, c_s)$ is **normalised** iff $d_1 \neq 0$.

Definition 12C. The **significant** digits of an FPN $\hat{x}$ are the digits following and including the first nonzero digit (in the mantissa).

Remark 12B. An FPN is normalised iff all digits are significant.

Theorem 12A. The largest (by absolute value) number in $\mathbb{R}_b(t,s)$ is
$$(.a \cdots a)_b \times b^{(a \cdots a)_b} = (.a \cdots a)_b \times b^M$$
where $a = b - 1$ and $M = b^s - 1$.

The smallest (i.e. closest to zero) number in $\mathbb{R}_b(t,s)$ is

- normalised $(.10\cdots0)_b \times b^{-M}$
- unormalised $(.0\cdots01)_b \times b^{-M}$

Definition 12D. **Overflow/underflow** occurs when the absolute value is too big/small.

Remark 12B. $\mathbb{R}_b(t,s)$ is finite whereas $\mathbb{R}$ is infinitely dense.

Definition 12E. Given $x \in \mathbb{R}$, the **representation** of x in $\mathbb{R}_b(s,t)$ is $\mathrm{fl}(x) = \hat{x}$ defined as

1. Normalise the mantissa:
$$\begin{aligned} x &= \pm(d_k d_{k-1}\cdots d_0.d_{-1}d_{-2}\cdots)_b \\ &= \pm(.D_1D_2\cdots)_b \times b^{k+1} \end{aligned}$$
2. *Chop/Round* the mantissa
   - *Chopping:* chop until (including) D_t.
   - *Rounding:* chop until D_t, roundup if $D_{t+1} \geqslant b/2$.
     Alternatively, add $b/2$ to $D_{\{t+1\}}$, then chop (cheaper)

1.3. Rounding Error & Error Propagation

Definition 13F. The (relative) **roundoff error** (RRO) between $x \in \mathbb{R}$ and its representation $\hat{x} \in \mathbb{R}_{b(t,s)}$ is
$$\delta_x = \frac{\hat{x} - x}{x}.$$
In other words, $\hat{x} = x(1 - \delta_x)$.

Theorem 13B. Given $\mathbb{R}_b(s,t)$, the relative roundoff error (RRO) δ is bounded by

- $0 \leqslant \delta < b^{1-t}$ for chopping (normalised)
- $|\delta| \leqslant \frac{1}{2}b^{1-t}$ for rounding (normalised)

Theorem 13B. Let $\mathsf{eps} \in \mathbb{R}_b(s,t)$ be the smallest nonnormalised float such that $1 + \mathsf{eps} > 1$, then

- $\mathsf{eps} = b^{1-t}$ for chopping
- $\mathsf{eps} = \frac{1}{2}b^{1-t}$ for rounding

Theorem 13C. The error propagation of multiplication is
$$\mathrm{fl}(\hat{x} \cdot \hat{y}) \approx (x \cdot y)(1 - \delta) \text{ where } |\delta| \leqslant 3\,\mathsf{eps}$$

Thus multiplication is *safe.*

Proof.

$$\begin{aligned}
\mathrm{fl}(\hat{x} \cdot \hat{y}) &= \mathrm{fl}(x(1-\delta_1) \cdot y(1-\delta_2)) \\
&= x(1-\delta_1) \cdot y(1-\delta_2) \cdot (1-\delta_3) \\
&= (x \cdot y)(1-\delta_1)(1-\delta_2)(1-\delta_3) \\
&= (x \cdot y)(1-\delta_1-\delta_2-\delta_3+\delta_1\delta_2+\delta_1\delta_3+\delta_2\delta_3-\delta_1\delta_2\delta_3) \\
&\approx (x \cdot y)(1-\delta_1-\delta_2-\delta_3) = (x \cdot y)(1-\delta)
\end{aligned}$$

where $\delta = \delta_1 + \delta_2 + \delta_3$, thus $|\delta| \leqslant 3\,\mathsf{eps}$. Note that the approximation is very good because

$$|\delta| < \mathsf{eps} \implies \delta^2 \ll \mathsf{eps} \text{ and } |\delta^3| \lll \mathsf{eps}.$$

□

Theorem 13D. The error propagation of addition is

$$\mathrm{fl}(\hat{x} + \hat{y}) \approx (x+y)(1-\delta) \text{ where } |\delta| \leqslant \begin{cases} 2\,\mathsf{eps} & \text{if } \operatorname{sgn} x = \operatorname{sgn} y \\ 2\,\mathsf{eps}\,\frac{|x-y|}{|x+y|} & \text{if } \operatorname{sgn} x \neq \operatorname{sgn} y \end{cases}$$

Thus addition is safe, but substraction is not safe and can causes *substractive cancellation.*

Proof.

$$\begin{aligned}
\mathrm{fl}(\hat{x} + \hat{y}) &= (x(1-\delta_1) + y(1-\delta_2))(1-\delta_3) \\
&= x(1-\delta_1)(1-\delta_3) + y(1-\delta_2)(1-\delta_3) \\
&\approx x(1-\delta_1-\delta_3) + y(1-\delta_2-\delta_3) && (\delta^2 \ll |\delta| < \mathsf{eps}) \\
&= (x+y)\left(1 - \frac{x}{x+y}(\delta_1+\delta_3) - \frac{y}{x+y}(\delta_2+\delta_3)\right) \\
&= (x+y)(1-\delta)
\end{aligned}$$

where

$$|\delta| \leqslant \left|\frac{x}{x+y}\right| 2\,\mathsf{eps} + \left|\frac{y}{x+y}\right| 2\,\mathsf{eps} = 2\frac{|x|+|y|}{|x+y|}\,\mathsf{eps}$$

□

1.4. Condition & Norm

Definition 14B. The **relative residual** of continuous $f : \mathbb{R} \to \mathbb{R}$ on $x \in \mathbb{R}$ is

$$\frac{|f(x) - f(\hat{x})|}{|f(x)|} = \frac{|x - \hat{x}|}{|x|} \operatorname{cond} f$$

where $\operatorname{cond} f$ is the **condition** of f, defined as

$$\operatorname{cond} f \stackrel{\text{def}}{=} \frac{|xf'(x)|}{|f(x)|}$$

Proof. Taylor expansion at $\hat{x}$:

$$f(x) \approx f(\hat{x}) + (x - \hat{x})f'(\hat{x})$$

□

Example 14C.

- $\operatorname{cond} \sqrt{x} = 1/2$ is bounded, so $\sqrt{x}$ is **well-conditioned**.
- $\operatorname{cond} \frac{c}{1-x^2} = \left|\frac{2x^2}{1-x^2}\right|$ is unbounded at $x = \pm 1$, so it's **ill-conditioned**.

Definition 14B. $\| - \| : V \to \mathbb{R}$ is a **norm** for vectorspace V iff

- $\|x\| \geqslant 0$ for any $x \in V$, and $\|x\| = 0 \iff x = 0$
- $\|\lambda x\| = |\lambda| \, \|x\|$
- $\|x + y\| \leqslant \|x\| + \|y\|$

Definition 14C. The vector **p-norm** of $\mathbb{R}^n$ is $\| - \|_p : \mathbb{R}^n \to \mathbb{R}$ defined as

$$\|\vec{x}\|_p \stackrel{\text{def}}{=} \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$$

Theorem 14E.

$$\|x\|_\infty \stackrel{\text{def}}{=} \lim_{p \to \infty} \|x\|_p = \max_i |x_i|$$

Definition 14D. The matrix **p-norm** of $\mathbb{R}^{n \times n}$ is $\| - \|_p :: \mathbb{R}^{n \times n} \to \mathbb{R}$ defined as

$$\|A\|_p \stackrel{\text{def}}{=} \max_{x \in \mathbb{R}^n} \frac{\|Ax\|_p}{\|x\|_p} = \max_{\|x\|_p = 1} \|Ax\|_p$$

Theorem 14B.

- $\|A\|_1$ is max sum of norm of col vectors

$$\|A_1\| = \max_j \sum_i |a_{ij}|$$

- $\|A\|_\infty$ is max sum of row vectors
$$\|A\|_\infty = \max_i \sum_j |a_{ij}| = \|A^T\|_1$$

1.5. Substraction Cancellation

Example 15B. (CSCC37 2023 Midterm) Write an algorithm that takes input $n \in \mathbb{N}$ and computes $\sum_{k=1}^{n} 1/k$. Your algorithm must be protected against performing unnecessary floating point operations in any floating point number system in which it is implemented.

Solution. Just compute until is less than machine epsilon.

```
sum = 0
for i in 1..n
  computer r as 1/k
  if sum + r is not greater than sum
    return sum
  sum = sum + r
return sum
```

When an expression $f(x)$ is evaluated using FP arithmetic, the *substraction cancellation* can cause poor results to be obtained for certain range of x. However, sometime an *alternate expression* can be used for such values of x. Note that if we are not given a specific FP system $\mathbb{R}_b(s,t)$, one can only determine an *approximate* range where the poor results are obtained.

Example 15B. (CSCC37 2023 Midterm)

- $(2-(2-x))$
  When $x \approx 0$, we have that $2 - x \approx 2$, so substractive cancellation can occur at $x \approx 0$, leading to poor results. An alternative expression is x, which has no subtraction, thus no subtractive cancellation at $x \approx 0$.

  Note that when $x \approx 2$, the expression $2 - x$ has subtractive cancellation, but $(2-(2-x))$ evaluates to approximately 2, so subtraction cancellation cannot happen for the original expresion.
- $1 - \sin x$
  Subtractive cancellation occurs at $x \approx 2k\pi + \pi/2$ because $\sin x$ evaluates to approximately 1. An alternative expression is

$$1 - \sin x = \frac{1 - \sin^2 x}{1 + \sin x} = \frac{\cos^2 x}{1 + \sin x}.$$

When $x \approx 2k\pi + \pi/2$, we have that $\sin x \approx 1$, so no subtractive cancellation for the alternative expression.

- $\sqrt{1+x} - \sqrt{1-x}$

  Note that when $x \approx 1$, subtractive cancellation can happen on $1 - x$, but not in the original expression because it approximates to $\sqrt{2}$.

  When $x \approx 0$, both $\sqrt{1+x}$ and $\sqrt{1-x}$ approximates to 1, so subtractive cancellation occurs. An alternative expression is

  $$\sqrt{1+x} - \sqrt{1-x} = \frac{1 + x - (1 - x)}{\sqrt{1+x} + \sqrt{1-x}} = \frac{2x}{\sqrt{1+x} + \sqrt{1-x}}$$

  which is fine for $x \approx 0$ as the denominator evaluates to approximately $1 + 1$ (addition on the same sign is safe).

- $e^x - 1$

  Subtractive cancellation occurs on $x \approx 0$ (causes addtion between different sign, similar magnitude). An alternative expression is (Tayler expansion)

  $$e^x - 1 = \sum_{1 \leqslant k < N} \frac{x^k}{k!}$$

  TODO for some large $N \in \mathbb{Z}^+$ (which?). (why no subtractive cancellation?)

2. Linear System of Equations

Definition 2E. A is **orthorgonal** iff $A^T = A^{-1}$.

Lemma 2C. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

2.1. Gaussian Elimination

Theorem 21B. (LU Factorisation) Let $A = (a_{ij}) \in \mathbb{R}^{n \times n}$, there exists *unit lower triangular* L and *upper triangular* U such that

$$A = LU.$$

Thus, $Ax = b$ can be solved by

1. Solve $Ld = b$ for d (forward substitution).
2. Solve $Ux = d$ for x (backward substitution).

Proof. Define L_1 as

$$L_1 = \begin{bmatrix} 1 & 0 & 0 \\ -a_{21}/a_{11} & 1 & 0 \\ -a_{31}/a_{11} & 0 & 1 \end{bmatrix}$$

Thus L_1 is the matrix that "eliminates" a_{21} and a_{31}. That is, $L_1 A$ is in the form

$$L_1 A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \tilde{a}_{22} & \tilde{a}_{23} \\ 0 & \tilde{a}_{32} & \tilde{a}_{33} \end{bmatrix}$$

Next, let L_2 (which eliminates $\tilde{a}_{32}$) be

$$L_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\tilde{a}_{32}/\tilde{a}_{22} & 1 \end{bmatrix}$$

We have that $U \stackrel{\text{def}}{=} L_2 L_1 A$ is upper triangluar. Finally, let $L \stackrel{\text{def}}{=} L_1^{-1} L_2^{-2}$, thus

$$A = LU$$

□

Lemma 21B. $L = L_1^{-1} + L_2^{-1} - I$

Lemma 21C. If $A = LU$, then $\det A = \det L$.

Proof. $\det : \mathbb{R}^{n \times n} \to \mathbb{R}$ is homomorphism. □

Remark 21C. $A = LU$ factorisation is unstable.

Proof. Let $\varepsilon = 10^{-16}$ be a very small number, and assume

$$L_1 A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & \varepsilon & 3 \\ 0 & 2 & 1 \end{bmatrix}$$

Thus, L_2 would be

$$L_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2/\varepsilon & 1 \end{bmatrix}$$

which causes an overflow. Furthermore, the result upper triangular U would also have overflow.

$$U \stackrel{\text{def}}{=} L_2 L_1 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2/\varepsilon & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & \varepsilon & 3 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & \varepsilon & 3 \\ 0 & 0 & 1 - 6/\varepsilon \end{bmatrix}$$

□

Example 21C.

$$\underbrace{\begin{bmatrix} 2 & -1 & 1 \\ 2 & 2 & 2 \\ -2 & 4 & 1 \end{bmatrix}}_{A} x = \underbrace{\begin{bmatrix} 1 \\ 8 \\ 7 \end{bmatrix}}_{b}$$

Solution.

$$\text{Let } L_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \implies L_1 A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & 1 \\ 0 & 3 & 2 \end{bmatrix}$$

$$\text{Let } L_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \implies L_2 L_1 A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix} = U$$

$$\therefore \text{Let } L = L_1^{-1} L_2^{-2} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

Thus $A = LU$, so $LUx = b$. Now, consider $Ld = b$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} d = \begin{bmatrix} 1 \\ 8 \\ 7 \end{bmatrix}$$

$$\begin{aligned} d_1 &= 1 \\ d_2 = 8 - d_1 &= 7 \\ d_3 = 7 + d_1 - d_2 &= 1 \end{aligned}$$

Finally, consider $Ux = b$

$$\begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix} x = \begin{bmatrix} 1 \\ 7 \\ 1 \end{bmatrix}$$

$$\begin{aligned} x_3 &= 1 \\ x_2 = (7 - x_3)/3 &= 2 \\ x_1 = (1 + x_2 - x_3)/2 &= 1 \end{aligned}$$

Thus $x = (1\ 2\ 1)^T$. (Notice it's the same process with pivoting)

2.1.1. Row Pivoting

Definition 211B. P_{ij} is the permutation matrix such that $P_{ij}A$ swaps row i and j in A, and AP_{ij} swaps column i and j.

Example 211B.

$$P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Remark 211B. $P_{ij}^2 = I$

Lemma 211D. Let $P_2 = P_{23}$, then

$$L_1 = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{bmatrix} \implies P_2 L_{P_2} = \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ a & 0 & 1 \end{bmatrix}$$

A better factorsation technique involves swapping rows such that the pivot has the largest norm. This ensures that each entry of L_k has norm bounded by 1, thus no overflow occurs.

Theorem 211B. (LU Factorisation with Pivot) Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$, there exists a *permutation* P, an *unit lower triangular* L, and an *upper triangular* U such that

$$PA = LU.$$

Thus, $Ax = b$ can be solved by $Ld = LUx = PAx = Pb$.

Proof. Assume, with pivot, we get

$$L_2 P_2 L_1 P_1 A = U$$

Thus let $\tilde{L}_1 = P_2 L_1 P_2$, i.e. L_1 with a_{21} and a_{31} swapped, it follows that

$$U = L_2 (P_2 L_1 P_2) P_2 P_1 A = L_2 \widetilde{L_1} P_2 P_1 A.$$

Thus, let $L = \widetilde{L_1}^{-1} L_2^{-1}$ and $P = P_2 P_1$, we have

$$PA = LU.$$

$\square$

Lemma 211C. If $PA = LU$, then $\det A = \det L / \det P$ where $\det P = \pm 1$ (i.e. the parity).

Example 211B.

$$\underbrace{\begin{bmatrix} 2 & 6 & 6 \\ 3 & 5 & 12 \\ 6 & 6 & 12 \end{bmatrix}}_{A} x = \underbrace{\begin{bmatrix} 20 \\ 25 \\ 30 \end{bmatrix}}_{b}$$

Solution.

$$\text{Let } P_1 = P_{13} \qquad \therefore P_1 A = \begin{bmatrix} 6 & 6 & 12 \\ 3 & 5 & 12 \\ 2 & 6 & 6 \end{bmatrix}$$

$$\text{Let } L_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ -1/3 & 0 & 1 \end{bmatrix} \qquad \therefore L_1 P_1 A = \begin{bmatrix} 6 & 6 & 12 \\ 0 & 2 & 6 \\ 0 & 4 & 2 \end{bmatrix}$$

$$\text{Let } P_2 = P_{23} \qquad \therefore P_2 L_1 P_1 A = \begin{bmatrix} 6 & 6 & 12 \\ 0 & 4 & 2 \\ 0 & 2 & 6 \end{bmatrix}$$

$$\text{Let } L_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1/2 & 1 \end{bmatrix} \qquad \therefore L_2 P_2 L_1 P_1 A = \begin{bmatrix} 6 & 6 & 12 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix} = U$$

$$\text{Let } \widetilde{L_1} = P_2 L_1 P_2 = \begin{bmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix} \therefore L = \widetilde{L_1}^{-1} L_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1/3 & 1 & 0 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

Thus, for $P = P_2 P_1 = P_{23} P_{13}$, it follows that $PA = LU$.

- Solve for d in $Ld = Pb$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1/3 & 1 & 0 \\ 1/2 & 1/2 & 1 \end{bmatrix} d = \begin{bmatrix} 30 \\ 20 \\ 25 \end{bmatrix}$$

$$\begin{aligned} d_1 &= 30 \\ d_2 = 20 - d_1/3 &= 10 \\ d_3 = 25 - d_2/2 - d_1/2 &= 5 \end{aligned}$$

- Solve for x in $Ux = d$

$$\begin{bmatrix} 6 & 6 & 12 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix} x = \begin{bmatrix} 30 \\ 10 \\ 5 \end{bmatrix}$$

$$\begin{aligned} x_3 &= 5/5 && = 1 \\ x_2 &= (10 - 2x_3)/4 && = 2 \\ x_1 &= (30 - 6x_2 - 12x_3)/6 && = 1 \end{aligned}$$

2.1.2. Numerical Singularity

Remark 212B. If at some stage k, the partial upper trigular matrix is 0 for $\hat{a}_{kk}, \hat{a}_{k+1,k}, \cdots$ (i.e. zero at kth column, below/include kth row), then the matrix U and A are **singular** (i.e. noninvertable). So we get zero or infinitely many solutions.

Definition 212B. A matrix is **near singular** iff at some stage k, all the elements include/below $\hat{a}_{kk}$ have magnitude bounded by

$$\mathsf{eps} \max_{1 \leqslant j \leqslant k} |u_{jj}|.$$

In that case, the matrix is said to have **numerical singularity**.

2.1.3. Complexity of Gaussian Elimination

We count the mult/add pairs (i.e. $mx + b$) as *flops* (floating point operation). Since the flop is the most expensive step (than comparasion), we are only counting flops.

Theorem 213D. (Complexity of Factorisation) In general, stage k takes $(n-k)^2$ operations, thus the complexity of $A = LU$ is

$$\sum_{k=1}^{n-1} (n-k)^2 = \sum_{k=1}^{n-1} k^2 = \frac{n(n-2)(2n-1)}{6} = \frac{n^3}{3} + O(n^2)$$

Remark 213B. At stage 1, we compute the matrix in the form

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & \hat{a}_{23} & \cdots & \hat{a}_{2n} \\ \vdots & \vdots & & \vdots \\ 0 & \hat{a}_{n2} & \dots & \hat{a}_{nn} \end{bmatrix}$$

which is $(n-1)^2$ operations.

Theorem 213B. (Complexity of Forward/Backward Substituion) Since d_k takes $k-1$ operations to solve, the complexity of $Ld = b$ is

$$\sum_{k=1}^{n} k - 1 = \frac{n^2}{2} + O(n)$$

Similarly, complexity of $Ux = d$ is $n^2/2 + O(n)$ as well.

Remark 213B. If $Ld = b$ where $L = (l_{ij})$ is unitary lower triangluar, then

$$\begin{aligned} d_1 &= b_1 && \text{(0 operation)} \\ d_2 &= \underline{b_2 - l_{21}d_1} && \text{(1 operation)} \\ d_3 &= \underline{\underline{b_3 - l_{32}d_2} - l_{31}d_1} && \text{(2 operations)} \\ &\vdots \end{aligned}$$

Remark 213C. Thus we only factorise A once, and solve different systems for different b.

2.1.4. Round Off Error

Definition 214B. Ideally, we want to solve $Ax = b$, but in reality, we get

$$(A + E)\hat{x} = b.$$

Let $r \stackrel{\text{def}}{=} E\hat{x}$ be the **residual**, then

$$(A + E)\hat{x} = b \iff r = b - A\hat{x}$$

Theorem 214C. In partial pivoting, for some not too large k that grows with n, we have that

- $\|E\| \lessapprox k \text{ eps } \|A\|$
- $\|r\| \lessapprox k \text{ eps } \|b\|$

Thus, the **relative residual** is bounded by

$$\frac{\|r\|}{\|b\|} \lessapprox k \text{ eps}$$

Definition 214B. The **condition** of matrix A is

$$\operatorname{cond}_p A \stackrel{\text{def}}{=} \|A\|_p \, \|A^{-1}\|_p.$$

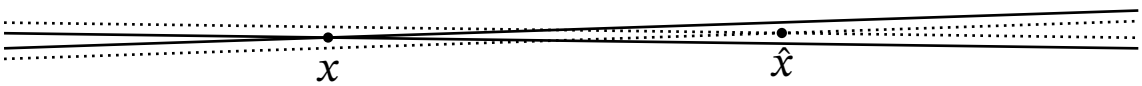
Notice $\operatorname{cond} A \geq 1$, and

$$\underbrace{\frac{\|x-\hat{x}\|}{\|x\|}}_{\substack{\text{relative}\\ \text{error}}} \leq \operatorname{cond} A \cdot \underbrace{\frac{\|r\|}{\|b\|}}_{\substack{\text{relative}\\ \text{residual}}}$$

Definition 214C.

1. If $\operatorname{cond} A \gg 1$, then the problem is **poorly** conditioned, meaning small relative residual does not imply small RRO.
2. If $\operatorname{cond} A \approx 1$, then the problem is **well** conditioned, meaning small relative residual is a good indicator of small RRO.
3. If $\operatorname{cond} A = 1$ (e.g. orthorgonal matrices), then the problem is **perfectly** conditioned.

The geometric interpretation of poorly conditioned is as follows:



2.2. Iterative Refinement

Algorithm 22C.

1. Compute $\hat{x}^{(1)}$ by solving $Ax = b$.
2. For each $i = 0, 1, \cdots$ until *converges* or *diverges*
   1. Compute $r^{(i)} = b - A\hat{x}^{(i)}$
   2. Solve $A\hat{z}^{(i)} = r^{(i)}$
   3. Let $\hat{x}^{(i+1)} = \hat{x}^{(i)} + \hat{z}^{(i)}$

Proof. Since we want to compute x such that $Ax = b$, but got $\hat{x}$ where $A\hat{x} = b - r$ where r is the residual, we now have

$$A\underbrace{(x-\hat{x})}_{z} = r$$

Therefore, we wish to solve the *residual system*, $Az = r$ and obtain $x = \hat{x} + z$, where z is the *correction*.

In reality, we get a computed $\hat{z}$, but $\hat{x} + \hat{z}$ is more precise than $\hat{x}$, thus we can continue the refinement. $\square$

3. Nonlinear Equation

3.1. Fixed Point Method

Definition 31D. To solve for nonlinear $F(x) = 0$, we consider a function g such that

$$F(x) = 0 \iff x = g(x)$$

Theorem 31B. (First-form) For any nonlinear F, let

$$g(x) = x - F(x)$$

be the **first form** fixpoint problem.

- x is root of F *iff* x is fixpoint of g.

Theorem 31B. (Second-form) Let arbitrary auxiliary function h, the **second form** fixpoint problem of F is

$$g(x) = x - h(x)F(x)$$

- x is root of F *only if* x is fixpoint of g.

Remark 31D. Second form gives more flexibility, and can make convergence faster.

3.1.1. Fixed Point Iteration

Algorithm 311B. (Fixed point iteration, FPI)

1. Define $\hat{x}_0$
2. Let $\hat{x}_{k+1} = g(\hat{x}_k)$ until *convergence*

- *X-test*: $|\hat{x}_{k+1} - x_k| < \varepsilon$
- *F-test*: $|f(\hat{x}_k)| < \varepsilon$

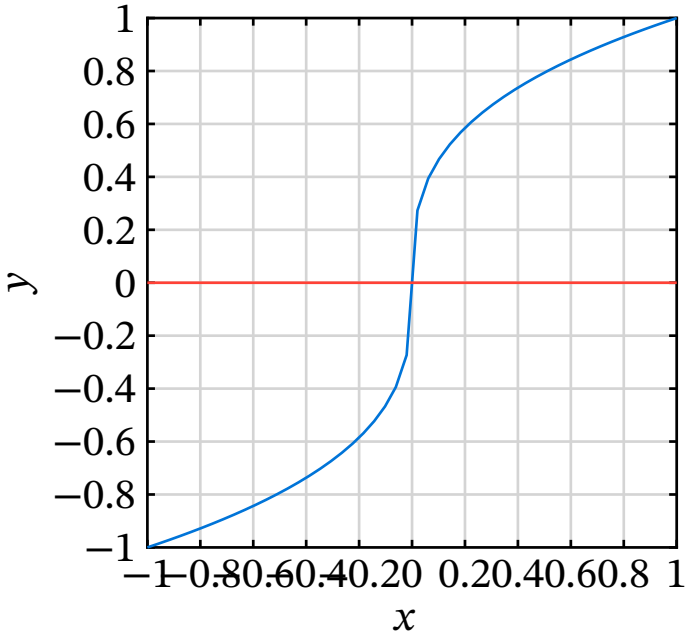


Figure 1: F-test fails, but X-test works

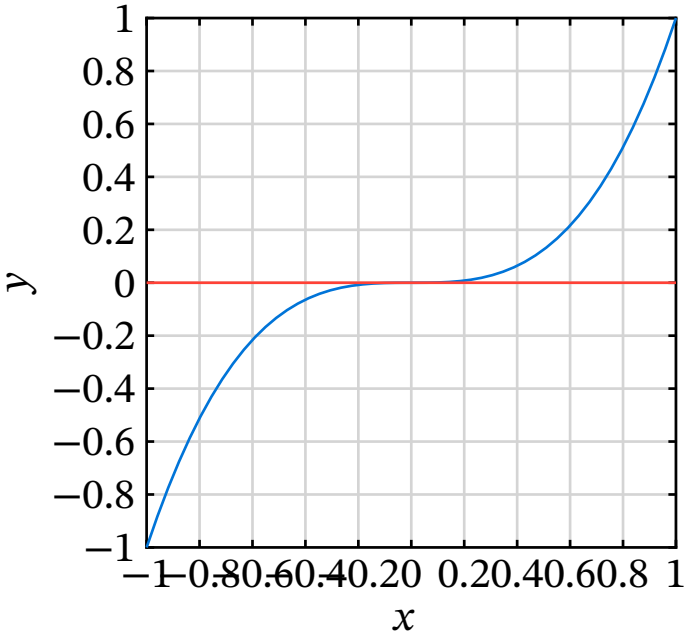


Figure 2: X-test fails, but F-test works

Theorem 311C. (Fixed point theorem, FPT) If there is interval $[a, b]$ satisfying $\exists L \forall x \in [a, b]$

1. (sink) $g(x) \in [a, b]$,
2. $|g'(x)| \leq L < 1$,

Then g has an unique fixed point in $[a, b]$.

3.1.2. Rate Of Convergence

Let $\tilde{x}$ be the fixpoint of g.

Definition 312B. FPI $x_{k+1} = g(x_k)$ has ***pth* order of convergence** iff

$$\lim_{x_k \to \tilde{x}} \frac{|\tilde{x} - x_{k+1}|}{|\tilde{x} - x_k|^p} = c \neq 0$$

Theorem 312B. (Rate of convergence theorem, RCT)

If

$$g'(\tilde{x}) = 0 \text{ and } \cdots \text{ and } g^{(p-1)}(\tilde{x}) = 0 \text{ and } g^{(p)}(\tilde{x}) \neq 0$$

Then the FPI $x_{k+1} = g(x_k)$ has exactly pth order of convergence.

- If $g^{(p)}(\tilde{x}) = 0$, then the FPT has *at least* pth order of convergence

3.2. Newton's Method

Algorithm 32B. (Newton's Method, NM)

$$x_{k+1} = x_k - \frac{F(x_k)}{F'(x_k)}$$

Notice this is FPI second form, with $h(x) = 1/F'(x_k)$

- It has at least second order of convergence, since $g''(\tilde{x}) = 0$

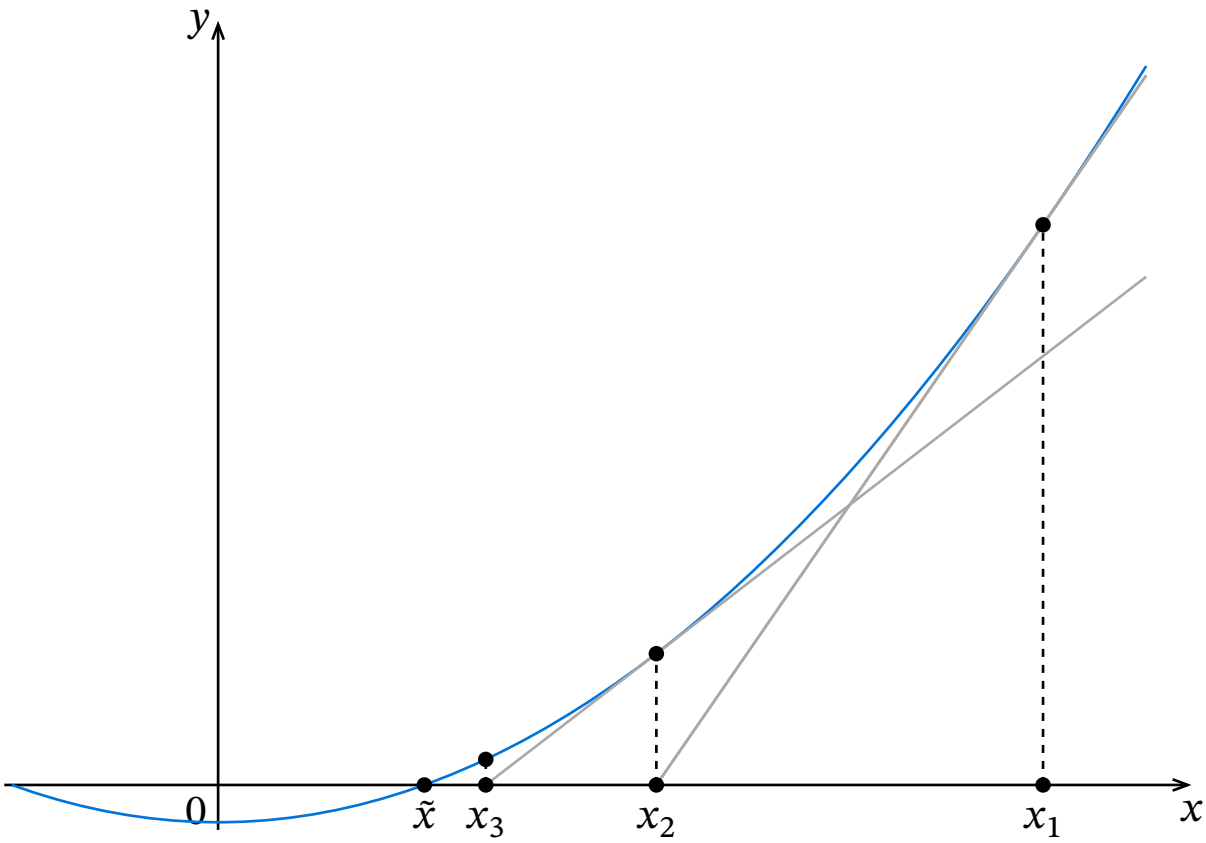


Figure 3: Geometric interpretation of NM

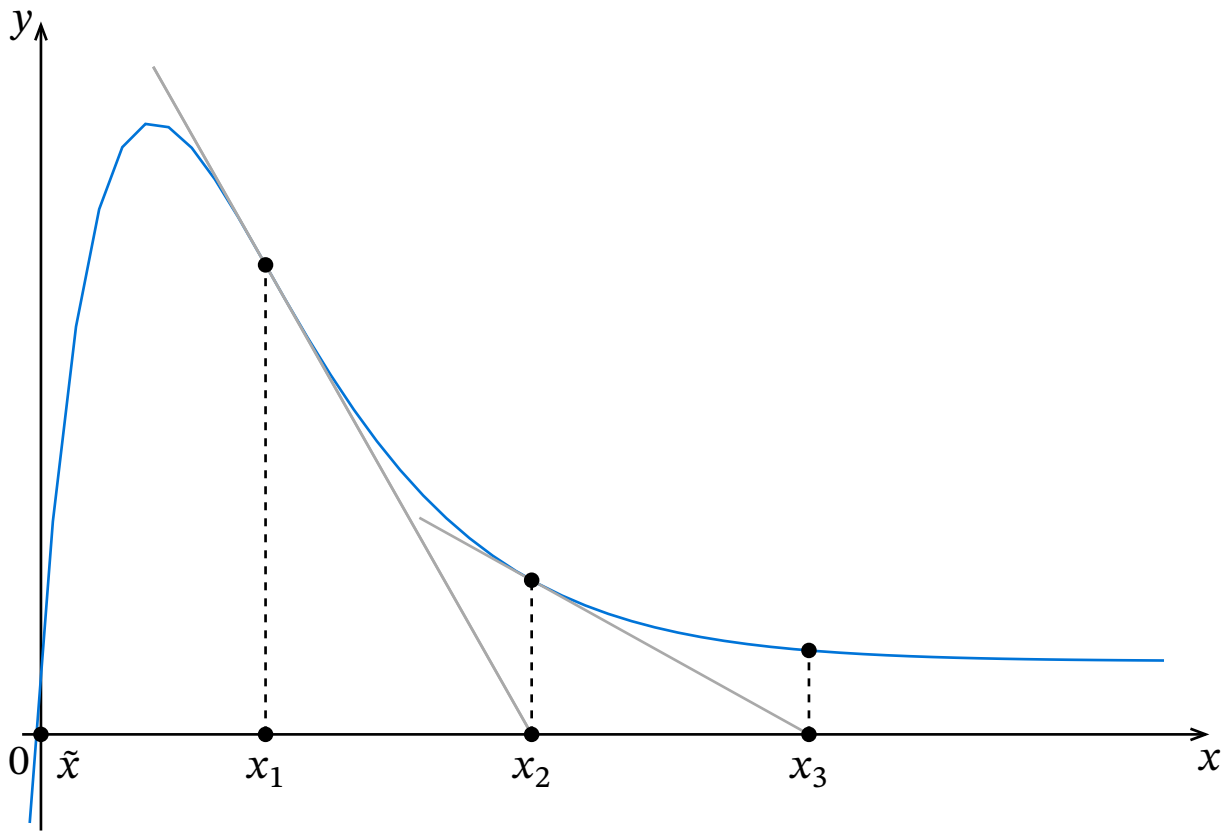


Figure 4: NM does not always converges.

3.3. Secant Method

NM but we approximate $f'(x)$ as

$$f'(x) \approx \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$$

Definition 33B. (Secant Method, SM)

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$$

Remark. SM is not directly FPI, so FPT/RCT cannot be applied. However, it has rate of convergence $p = \frac{1+\sqrt{5}}{2} \approx 1.62$

Theorem 33B. SM has lower rate of convergence as NM, but NM requires two function evaluation on each iteration (f and f') whereas SM requires only one (i.e. f). Therefore, SM is **effectively faster** than NM.

3.4. Bisection Method

Binary search.

Definition 34B.

1. Choose interval $[a_0, b_0]$ such that $f(a_0) \leq 0 \leq f(b_0)$
2. Loop until $b_k - a_k \leq \varepsilon$
    1. Let $m = (a_k + b_k)/2$
    2. If $f(m) \leq 0$ then
        1. $(a_{k+1}, b_{k+1}) = (m, b_k)$
    3. else
        1. $(a_{k+1}, b_{k+1}) = (a_k, m)$

Theorem 34B. BM

- will always converge (without FPT)
- has rate of convergence $p = 1$ (with $c = 1/2$)

3.5. Hybrid

combine NM and BM

4. Approximation & Interpolation

4.1. Types of Approximation

Definition 41B. (Truncated Taylor series) Given differentiable function $f : D \subseteq \mathbb{R} \to \mathbb{R}$ and fix point $a \in D$, the **truncated Taylor series** of f is the approximation

$$p(x) = f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}}{n!}(x-a)^n = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

The **error** is

$$e(x) \stackrel{\text{def}}{=} f(x) - p(x) = \frac{f^{(n+1)}(\eta)}{(n+1)!}(x-a)^{n+1}$$

for some $\eta \in [x, a]$

Definition 41B. (Types of approximation)

Let f be the underlying function to approximate

1. *Interpolation.*
   Find $p(x)$ such that $p(x_i) = f(x_i)$ for each $0 \le i \le n$
2. *Least Squares Approximation.*
   Find $p(x)$ to minimise $\|f - p\|_2$ where

   $$\|f - p\|_2 \stackrel{\text{def}}{=} \left(\int_a^b (f(x) - p(x)^2) dx \right)^{1/2}$$

   Other norms:
   - min-max approximation

     $$\|f - p\|_\infty \stackrel{\text{def}}{=} \max_{a \le x \le b} |f(x) - p(x)|$$

   - 1-norm approximation

     $$\|f - p\|_1 \stackrel{\text{def}}{=} \int_a^b |f(x) - p(x)| \, dx$$

4.2. Polynomial Approximation

Definition 42C. Let $\mathcal{P}_n \stackrel{\text{def}}{=} \{f(x) \in \mathbb{R}[x] : \deg f(x) \le n\}$ be polynomials of degree at most n.

Remark. $\mathcal{P}_n$ is a function space with dimention $n+1$.

Remark. $\mathcal{P}_n$ can be spanned by the **monomial basis** $\{(x-c)^i\}_{i \le n}$ for any c.

Theorem 42B. (Weierstrass' theorem) If $f \in C[a,b]$, then for any $\varepsilon > 0$

$$\exists p \in \mathcal{P}_n \quad \|f - p\|_\infty < \varepsilon$$

4.3. Numerical Methods for Polynomial Interpolation

4.3.1. Vandermonde (Undetermined Coefficients)

Theorem 431B. Given function $\{(x_i, y_i)\}_{0 \le i \le n}$, there exists an unique $p(x) \in \mathcal{P}_n$ such that $p(x_i) = y_i$ for each i.

Proof. We wish to find coefficients $\boldsymbol{a} \in \mathbb{R}^n$ satisfying

$$p(x_i) \stackrel{\text{def}}{=} a_0 + a_1 x_i + \cdots + a_n x_i^n = y_i.$$

Thus, we set up the system $V\boldsymbol{a} = \boldsymbol{y}$, where V is the **Vandermonde matrix**

$$\left[\begin{array}{ccccc|c} 1 & x_0 & x_0^2 & \cdots & x_0^n & y_0 \\ 1 & x_1 & x_1^2 & \cdots & x_1^n & y_1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n & y_n \end{array}\right]$$

□

Lemma 431B. V is invertible

Proof. Observe that V is invertible iff V is nonsingular, meaning $\nexists a \neq 0$ such that $Va = 0$.

BWOC let $a \neq 0$ be root of V. Thus (by definition of V) each distinct x_i is root of $p \in \mathcal{P}_n$, contradicting that p only has at most n roots (since $\deg p \le n$ and FToAlgebra). □

Lemma 431C. The determinant of Vandermonde matrix is

$$\det V = (-1)^{n+1} \prod_{i=1}^{n} \prod_{j=0}^{i-1} (x_i - x_j) \neq 0$$

Remark 431B. The quality of Vandermonde depends on condition of V, and cost of $PA = LU$ (i.e. $n^3/3 + O(n^2)$ flops). Typically very bad.

4.4. Lagrange Rules

Theorem 44D. (Lagrange basis) Given data points $\{(x_i, y_i)\}$, the **lagrange basis** is, for $0 \le i \le n$

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j} \in \mathcal{P}_n.$$

The **langrage interpolation** is

$$p(x) = \sum_{0 \le i \le n} y_i \ell_i(x)$$

satisfying $p(x_i) = y_i$ for each i

Lemma 44B. For each $0 \le i, j \le n$

$$\ell_i(x_j) = \begin{cases} 1 & j = i \\ 0 & j \neq i \end{cases}$$

Remark 44B. Lagrange polynomial is **free** to construct, but **prohibitively expensive** to evaluate at non-interpolation (i.e. between) points.

4.5. Newton Basis (Divided Differences)

Theorem 45C. To find $p(x) \in \mathcal{P}_n$ satisfying $p(x_i) = y_i$, we seek $p(x)$ in the form

$p(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n(x - x_0)\cdots(x - x_{n-1})$

Which is equivalent to forward solving

$$\left[\begin{array}{cccccc|c} 1 & 0 & \cdots & \cdots & \cdots & 0 & y_0 \\ 1 & x_1 - x_0 & 0 & \cdots & \cdots & 0 & y_1 \\ 1 & x_2 - x_0 & (x_2 - x_0)(x_2 - x_1) & 0 & \cdots & 0 & y_2 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & x_n - x_0 & \cdots & & & & y_n \end{array}\right]$$

4.5.1. Divided Differences

Definition 451B. (Divided difference)

$$y[x_i] = y_i$$

$$y[x_{i+k}, \cdots, x_i] = \frac{y[x_{i+k}, \cdots, x_{i+1}] - y[x_{i+k-1}, \cdots, x_i]}{x_{i+k} - x_i}$$

Example 451C.

$$y[x_2, x_1, x_0] = \frac{y[x_2, x_1] - y[x_1, x_0]}{x_2 - x_0}$$
$$= \frac{\frac{y_2 - y_1}{x_2 - x_1} - \frac{y_1 - y_0}{x_1 - x_0}}{x_2 - x_0}$$

Theorem 451B. (Newton polynomial) Let

$$p(x) = y[x_0] + (x - x_0)y[x_1, x_0] + \cdots + (x - x_0)\cdots(x - x_{n-1})y[x_n, \cdots, x_0]$$
$$= \sum_{0 \le k \le n} \left(\prod_{0 \le j \le k} (x - x_j) \right) y[x_k, \cdots, x_0]$$

then $p(x) \in \mathcal{P}_n$ satisfying $p(x_i) = y_i$

Example 451B. Find $p(x) \in \mathcal{P}_3$ such that

$$\{(0, 1), (1, 3), (2, 9), (3, 25)\} \subseteq p$$

Proof. Consider the **divided difference table**

x	$y[x_i]$	$y[x_{i+1}, x_i]$	$y[x_{i+2}, \cdots, x_i]$	$y[x_{i+3}, \cdots, x_i]$
0	$\boxed{1}$			
		$\frac{3-1}{1-0} = \boxed{2}$		
1	3		$\frac{6-2}{2-0} = \boxed{2}$	
		$\frac{9-3}{2-1} = 6$		$\frac{5-2}{3-0} = \boxed{1}$
2	9		$\frac{16-6}{3-1} = 5$	
		$\frac{25-9}{3-2} = 16$		
3	25			

Thus

$$p(x) = 1 + 2x + 2x(x - 1) + 1x(x - 1)(x - 2)$$

□

4.5.2. Osculatory Interpolation (with Derivative Data)

Lemma 452B.

$$\lim_{x_1 \to x_0} y[x_1, x_0] \stackrel{\text{def}}{=} \lim_{x_1 \to x_0} \frac{y_1 - y_0}{x_1 - x_0} = y'(x_0)$$

In general,

$$\lim_{\substack{x_k \to x_0 \\ \vdots \\ x_1 \to x_0}} y[x_k, \cdots, x_0] = \frac{y^{(k)}(x_0)}{k!}$$

Example 452C. Find $p \in \mathcal{P}_4$ such that

$$p(0) = 0 \quad p(1) = 1 \quad p(2) = 6$$
$$p'(1) = 1$$
$$p''(1) = 2$$

Proof. Consider the divided difference table

x_i	$y[x_i]$	$y[x_{i+1}, x_i]$	$y[x_{i+2}, \cdots, x_i]$	$y[x_{i+3}, \cdots, x_i]$	$y[x_{i+4}, \cdots, x_i]$
0	0				
		$\frac{1-0}{1-0} = 1$			
1	1		$\frac{1-1}{1-0} = 0$		
		$\frac{y'(1)}{1!} = 1$		$\frac{1-0}{1-0} = 1$	
1	1		$\frac{y''(1)}{2!} = 1$		$\frac{3-1}{2-0} = 1$
		$\frac{y'(1)}{1!} = 1$		$\frac{4-1}{2-1} = 3$	
1	1		$\frac{5-1}{2-1} = 4$		
		$\frac{6-1}{2-1} = 5$			
2	6				

Thus,

$$p(x) = x + x(x-1)^2 + x(x-1)^3$$

□

4.6. Error in Polynomial Interpolation

Definition 46B.

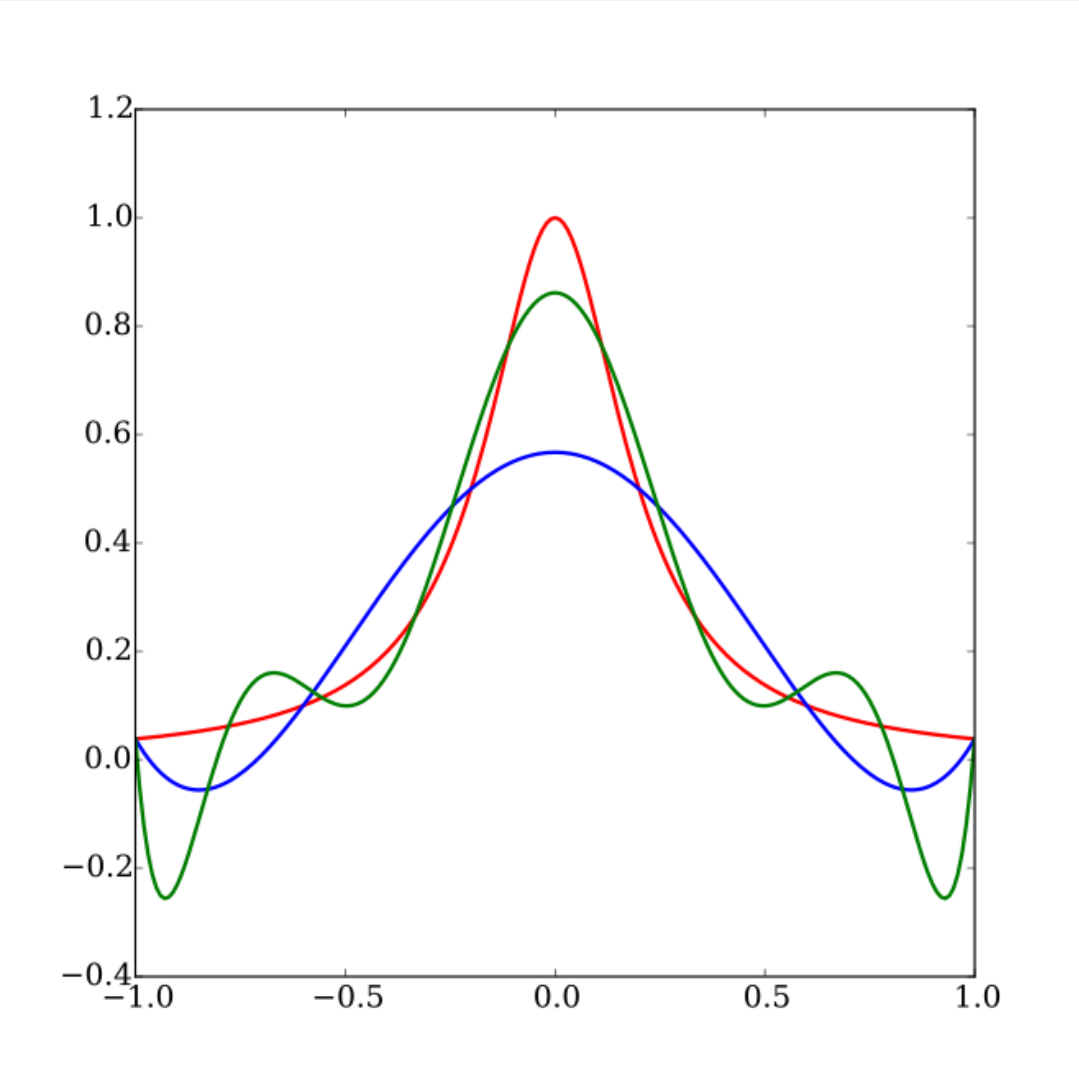
$$E(x) = y(x) - p(x)$$

Theorem 46B.

$$E(x) = \frac{y^{(n+1)}}{(n+1)!}(\eta) \prod_{0 \le i \le n} (x - x_i)$$

for some $\eta \in \operatorname{span}\{x_0, \cdots, x_n, x\} = \left[\min_{\{x_0,\cdots,x_n,x\}}, \max_{\{x_0,\cdots,x_n,x\}}\right]$

Example 46B. (Runge phenomenon)



Going to higher degree does not necessary improve accuracy

- $f(x) = 1/(1 + 25x^2)$ is the underlying function
- $p_5(x)$ is 5th order approximation (agrees with $f(x)$ at 6 points)
- $p_9(x)$ is 9th order approximation (agrees with $f(x)$ at 10 points)

Notice that p_9 is a worse approximation than p_5

- Reason, $E(x)$ blows up faster as derivate gets higher

5. Numerical Integration

5.1. Quadrature

We wish to compute

$$I(f) \stackrel{\text{def}}{=} \int_a^b f(x)dx \approx I(p)$$

Definition 51B. (Standard quadrature formula) Given $A_i : \mathbb{R} \to \mathbb{R}$ **weights** and $x_i \in \mathbb{R}$ nodes, the **interpolation quadrature** for $f(x)$ is

$$Q(f) \stackrel{\text{def}}{=} I(p) = \sum_{i=0}^{n} A_i f(x_i)$$

Example 51B. If we approximate $f(x)$ in *Lagrange form*, as

$$p(x) = \sum_{i=0}^{n} f(x_i)\ell_i(x)$$

Then the interpolation quadrature is

$$\begin{aligned} Q(f) = I(p) &= \int_a^b \sum_{i=0}^{n} f(x_i)\ell_i(x)dx \\ &= \sum_{i=0}^{n}\left(\int_a^b \ell_i(x)dx\right)f(x_i) = \sum_{i=0}^{n} A_i f(x_i) \end{aligned}$$

where

$$A_i(x) = \int_a^b \ell_i(x)dx$$

Lemma 51B. Q is a linear operator, i.e.

$$Q(\alpha f + g) = \alpha Q(f) + Q(g)$$

Thus if $Q(f)$ integrates $1, x, \cdots, x^n$ exactly, then Q integrates every $p(x) \in \mathcal{P}_n$ exactly.

Theorem 51B. (Existence of interpolating quadrature rules) Given any distinct nodes $\{x_0, \cdots, x_n\}$, there exists *unqiue* weights A_i such that the $Q(f)$ integrates $\mathcal{P}_n$ exactly.

Proof. We wish to find A_i such that for each k,

$$Q(x^k) \stackrel{\text{def}}{=} \sum_{i=0}^{n} A_i x_i^k = \int_a^b x^k dx = \frac{b^{k+1} - a^{k+1}}{k+1}$$

which is equivalent to solving $V^T \boldsymbol{A} = \boldsymbol{b}$

$$\left[\begin{array}{cccc|c} 1 & 1 & \cdots & 1 & b-a \\ x_0 & x_1 & \cdots & x_n & (b^2-a^2)/2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_0^n & x_1^n & \cdots & x_n^n & (b^{n+1}-a^{n+1})/(n+1) \end{array}\right]$$

where V^T is transposed Vandermonde matrix (i.e. invertible), thus solution exists and is unique. $\square$

5.2. Precision

Definition 52B. (Precision) The **presion** of Q is the largest $m \in \mathbb{N}$ such that Q integrates every $p(x) \in \mathcal{P}_m$ exactly.

Theorem 52C. The **midpoint rule** ($n = 0$), defined as

$$I(f) \approx M(f) = I(p_0) \stackrel{\text{def}}{=} (b - a) f\left(\frac{a+b}{2}\right)$$

has precision $m = 1$.

Proof.

$$M(1) = b - a = I(1)$$

$$M(x) = (b-a)\frac{a+b}{2} = \frac{b^2 - a^2}{2} = I(x)$$

$$M(x^2) = (b-a)\left(\frac{a+b}{2}\right)^2 \neq \frac{b^3 - a^3}{3} = I(x^2)$$

TODO figure $\square$

Theorem 52B. The **trapezoidal rule** ($n = 1$), defined as

$$I(f) \approx T(f) = I(p_1) \stackrel{\text{def}}{=} \frac{b-a}{2}[f(a) + f(b)]$$

has precision $m = 1$

TODO figure

Theorem 52C. The **Simpson's rule** ($n = 2$), defined as

$$I(f) \approx S(f) = I(p_2) \stackrel{\text{def}}{=} \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

5.3. Error in Interpolating Quadrature

Definition 53B. THe **error** is

$$e_n \stackrel{\text{def}}{=} I(f) - Q_n(f) = \int_a^b [f(x) - p_n(x)]dx$$
$$= \int_a^b \left[\frac{f^{(n+1)}(\eta)}{(n+1)!} \prod_{i=0}^{n} (x - x_i) \right] dx$$

Theorem 53D. Let $f \in C[a,b]$ and $Q_n(f) = \sum_{i=0}^{n} A_i^{(n)} f\left(x_i^{(n)}\right)$, then $\lim_{n\to\infty} Q_n(f) = I(f)$ if and only if

$$\exists k \forall n \quad \sum_{i=0}^{n} \left| A_i^{(n)} \right| \leq k$$